Classification of Acute Lymphoblastic Leukemia

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BASELINE MODEL: Random Forest directly trained on pixels values

```
Accuracy: 0.8680981595092024

Confusion Matrix:
[[ 50  32   1  18]
 [  9 188   0   0]
 [  6  19 167   1]
 [  0   0   0 161]]

Classification Report:
              precision    recall  f1-score   support

           0       0.77      0.50      0.60       101
           1       0.79      0.95      0.86       197
           2       0.99      0.87      0.93       193
           3       0.89      1.00      0.94       161

    accuracy                           0.87       652
   macro avg       0.86      0.83      0.83       652
weighted avg       0.87      0.87      0.86       652
```

Medically Relevant Features

```
Best Model (LightGBM) Accuracy: 0.9601 (96.01%)
Training Time: 4.79 seconds
Confusion Matrix:
[[ 92   4   1   4]
 [  3 188   2   4]
 [  1   1 189   2]
 [  0   4   0 157]]
```

GOOD!! ✅

```
--------------------------------------------------------------
TRAINING LIGHTGBM WITH ADVANCED FEATURES
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Training time: 0.957 seconds

==================================================
ACCURACY: 0.9969 (99.69%)
==================================================

Classification Report:
              precision    recall  f1-score   support

           0       0.98      1.00      0.99       101
           1       1.00      0.99      1.00       197
           2       1.00      1.00      1.00       193
           3       1.00      0.99      1.00       161

    accuracy                           1.00       652
   macro avg       1.00      1.00      1.00       652
weighted avg       1.00      1.00      1.00       652


Confusion Matrix:
[[101   0   0   0]
 [  1 196   0   0]
 [  0   0 193   0]
 [  1   0   0 160]]
```
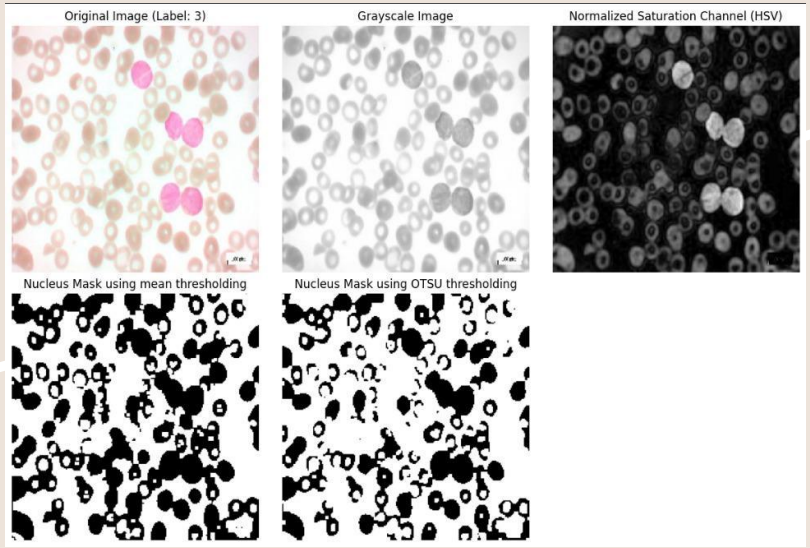

BAD!!! X

Complex Features

Feature Extraction:

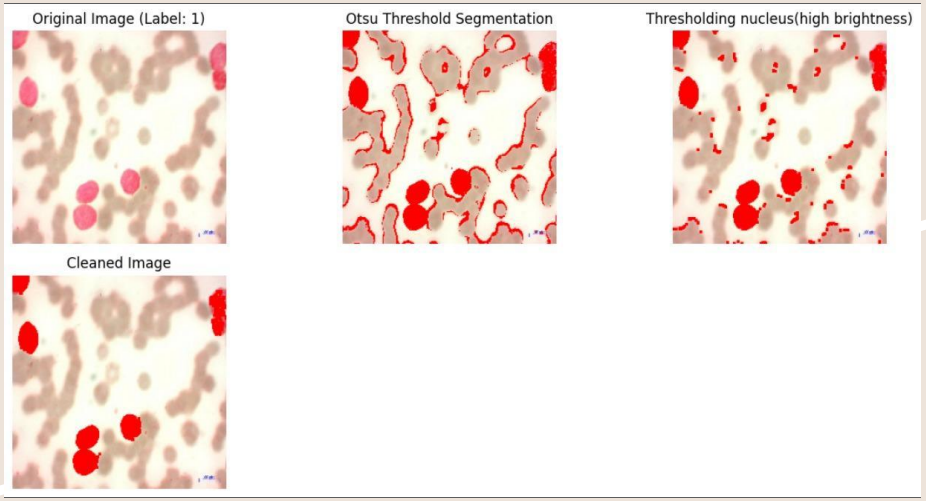
- Manual Feature Extraction
- 22 features extracted that was cut down to 18
- Nucleus - Cytoplasm features and Shape features are the main features of our model



	feature	importance
1	Cytoplasm_Pixels	19021
19	Edge_Mean	14756
5	CytoMean	14202
0	Nucleus_Pixels	13373
4	CytoNuc_Area_Ratio	9391
16	Circularity	7517
9	Perimeter	7394
13	MajorAxisLength	7370
6	NucMean	7045
17	ConvexArea	5697
11	Solidity	5682
14	MinorAxisLength	5076
8	Area	4940
7	CytoNuc_MeanDiff	4820
10	Eccentricity	4442
21	Nuc_vs_Cyto_MeanRatio	4138
18	Convexity	3612
12	Extent	3148

Main Challenge & Models Comparison:

Rank	Model	Accuracy
1	LightGBM	95.71% <= BEST!
2	XGBoost	95.25%
3	Gradient Boosting	94.79%
4	Random Forest (500 trees)	94.02%
5	Random Forest (250 trees)	93.87%
6	K-Nearest Neighbors (k=5)	90.64%
7	K-Nearest Neighbors (k=10)	90.64%
8	Decision Tree	90.49%
9	AdaBoost	76.99%
10	SVM (RBF kernel)	30.37%
11	Naive Bayes	29.91%
12	Logistic Regression	26.99%



FINE TUNING THE BEST MODEL

```
num_leaves=32, learning_rate=0.04, n_estimators=1500 → Accuracy: 0.9586, Time: 1.93s
num_leaves=64, learning_rate=0.01, n_estimators=500 → Accuracy: 0.9586, Time: 3.22s
num_leaves=64, learning_rate=0.01, n_estimators=800 → Accuracy: 0.9571, Time: 5.73s
num_leaves=64, learning_rate=0.01, n_estimators=1000 → Accuracy: 0.9571, Time: 5.12s
num_leaves=64, learning_rate=0.01, n_estimators=1200 → Accuracy: 0.9586, Time: 6.99s
num_leaves=64, learning_rate=0.01, n_estimators=1500 → Accuracy: 0.9586, Time: 6.69s
num_leaves=64, learning_rate=0.02, n_estimators=500 → Accuracy: 0.9586, Time: 2.36s
num_leaves=64, learning_rate=0.02, n_estimators=800 → Accuracy: 0.9601, Time: 3.26s
num_leaves=64, learning_rate=0.02, n_estimators=1000 → Accuracy: 0.9586, Time: 3.18s
num_leaves=64, learning_rate=0.02, n_estimators=1200 → Accuracy: 0.9586, Time: 3.80s
num_leaves=64, learning_rate=0.02, n_estimators=1500 → Accuracy: 0.9586, Time: 4.39s
```

The PBS images have RBCs and different types of WBCs. We had to identify RBCs and WBCs and identify the lymphoblasts with clear nucleus and cytoplasm.

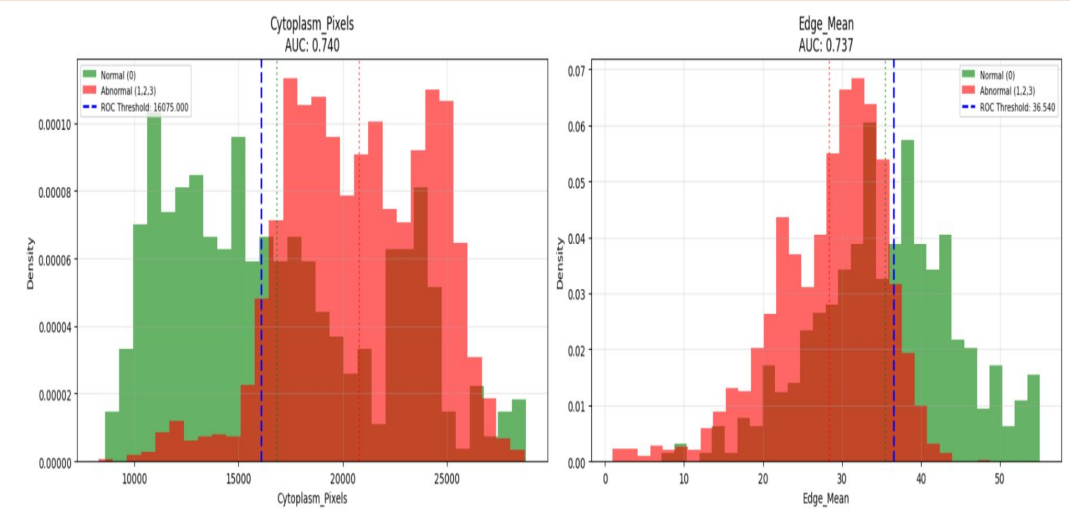
Thresholding

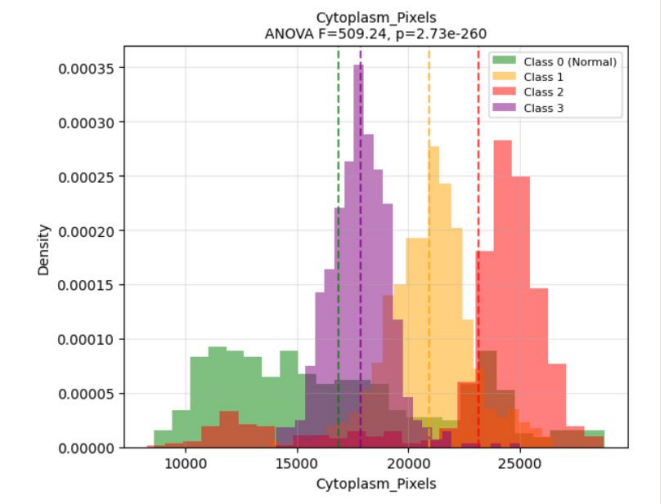
Normal vs Abnormal

- **ROC Curve Analysis**
- **Distribution Analysis**
- **Decision Tree Split Points**

Multi-Class Thresholding

- **One vs Rest ROC Analysis**
- **Statistical Range Analysis**
- **Decision Boundary Ranges**





Results:

Our final model was an IF-ELSE MODEL!

```
================================================================================
COMPARISON: Different Threshold Methods
================================================================================

ROC-Optimized Thresholds:
  Accuracy:    90.0%
  Sensitivity: 91.1%
  Specificity: 84.2%
  Precision:   96.9%
  F1 Score:    0.939

Decision Tree Thresholds:
  Accuracy:    91.7%
  Sensitivity: 95.3%
  Specificity: 72.3%
  Precision:   94.9%
  F1 Score:    0.951

Statistical (IQR) Thresholds:
  Accuracy:    61.2%
  Sensitivity: 54.8%
  Specificity: 96.0%
  Precision:   98.7%
  F1 Score:    0.705

================================================================================
COMPARISON: Threshold Classifiers vs Best ML Model
================================================================================

Method                    Accuracy     Sensitivity  Specificity  F1 Score
--------------------------------------------------------------------------------
Decision Tree              91.7%        95.3%        72.3%        0.951
ROC-Optimized              90.0%        91.1%        84.2%        0.939
Statistical (IQR)          61.2%        54.8%        96.0%        0.705

Best ML Model (LGBM):  98.0%
```

```python
def classify_with_roc_thresholds(sample):
    """ROC-optimized thresholds (maximize sensitivity + specificity)"""
    votes_abnormal = 0
    total_rules = 0

    if 'Cytoplasm_Pixels' in sample:
        if sample['Cytoplasm_Pixels'] >= 16075.0:
            votes_abnormal += 3
        total_rules += 3

    if 'Edge_Mean' in sample:
        if sample['Edge_Mean'] <= 36.5404:
            votes_abnormal += 3
        total_rules += 3

    if 'Nucleus_Pixels' in sample:
        if sample['Nucleus_Pixels'] >= 4728.0:
            votes_abnormal += 2
        total_rules += 2

    if 'CytoNuc_Area_Ratio' in sample:
        if sample['CytoNuc_Area_Ratio'] <= 2.4479:
            votes_abnormal += 2
        total_rules += 2

    if total_rules == 0:
        return 0
    return 1 if votes_abnormal / total_rules > 0.5 else 0
```

	Feature	Train_AUC	Test_Accuracy	Test_Sensitivity	Test_Specificity	Test_Precision	Threshold	Direction
0	Cytoplasm_Pixels	0.740	86.2%	95.5%	35.6%	89.0%	16075.0000	high
1	Edge_Mean	0.737	87.7%	92.6%	61.4%	92.9%	36.5404	low
2	Nucleus_Pixels	0.709	46.9%	38.7%	92.1%	96.4%	4728.0000	high
3	CytoNuc_Area_Ratio	0.639	36.3%	25.0%	98.0%	98.6%	2.4479	low

THANK YOU!

Contributions:
Kshetrimayum Paresh Singh and Abhishek Nair: Worked on Feature Extraction
Kushagra: Worked on Model building and interpretation